

FARM MANAGEMENT SYSTEM

User Story:

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I want to use a centralized Salesforce-based platform to record and manage farm, crop, farmer, and buyer information so that all agricultural data is stored securely and systematically. I want to enter and update farm details, crop production records, and buyer transactions easily through a user-friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive information is protected while enabling smooth collaboration among users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management.

1. INTRODUCTION

1.1 Project Overview :

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

1.2 Purpose :

The main purpose of this project is to:

- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring
- Support data-driven farming decisions
- Enhance productivity and transparency

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	Frustrated and confused
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient
PS-3	Buyer	Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS-4	System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure

Problem Statements – Farm Management System

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1		Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	
PS-2	 Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient 
PS-3		Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied 
PS-4	 System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure 

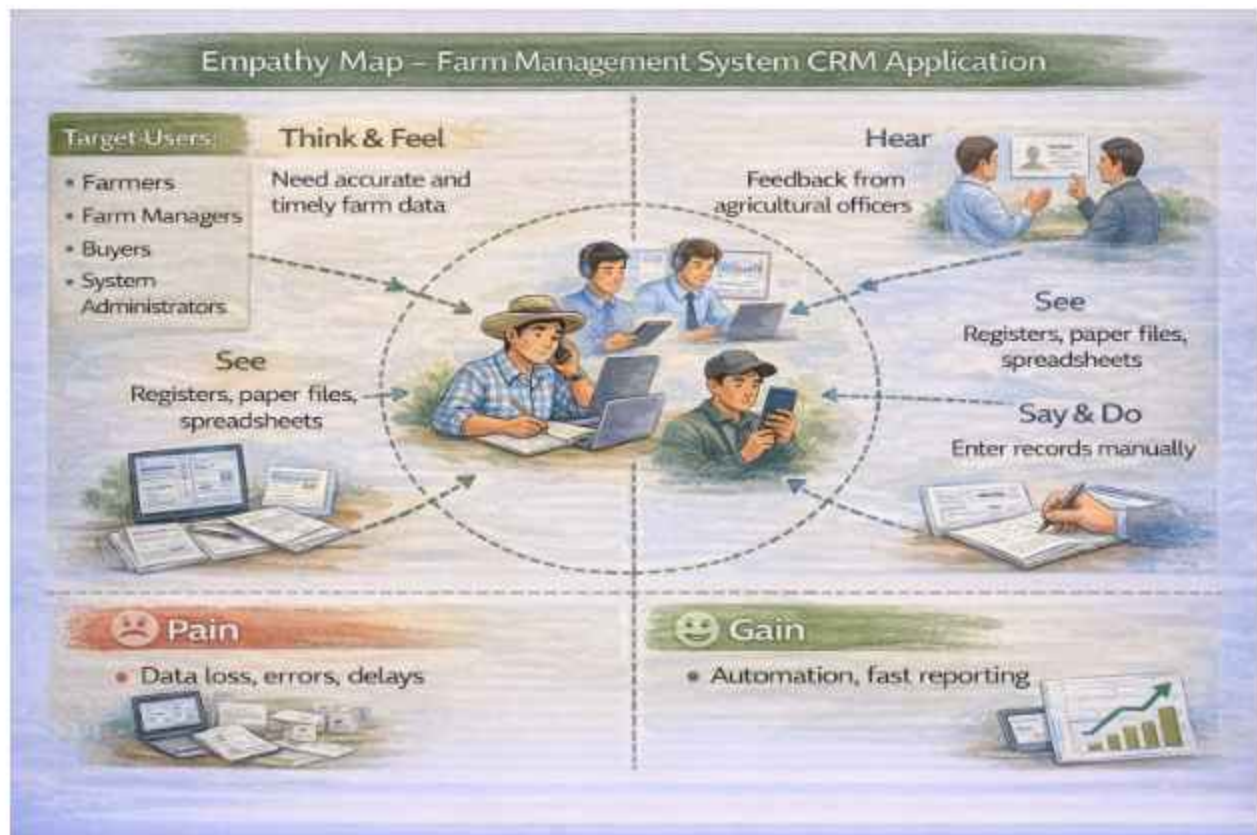
2.2 Empathy Map Canvas

Target Users:

- Farmers
- Farm Managers
- Buyers
- System Administrators

Empathy Analysis

- **Think & Feel:** Need accurate and timely farm data
- **Hear:** Feedback from agricultural officers
- **See:** Registers, paper files, spreadsheets
- **Say & Do:** Enter records manually
- **Pain:** Data loss, errors, delays
- **Gain:** Automation, fast reporting



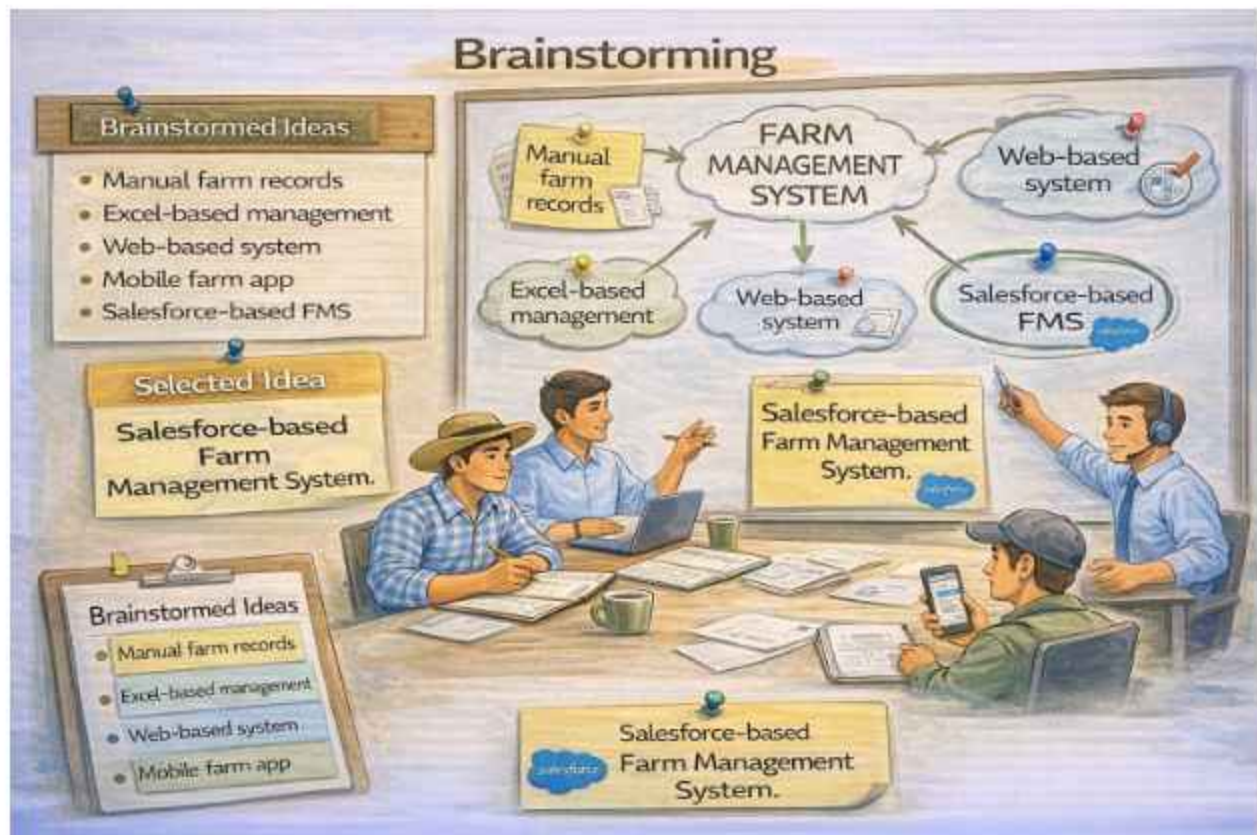
2.3 Brainstorming

Brainstormed Ideas

- Manual farm records
- Excel-based management
- Web-based system
- Mobile farm app
- Salesforce-based FMS

Selected Idea

Salesforce-based Farm Management System.

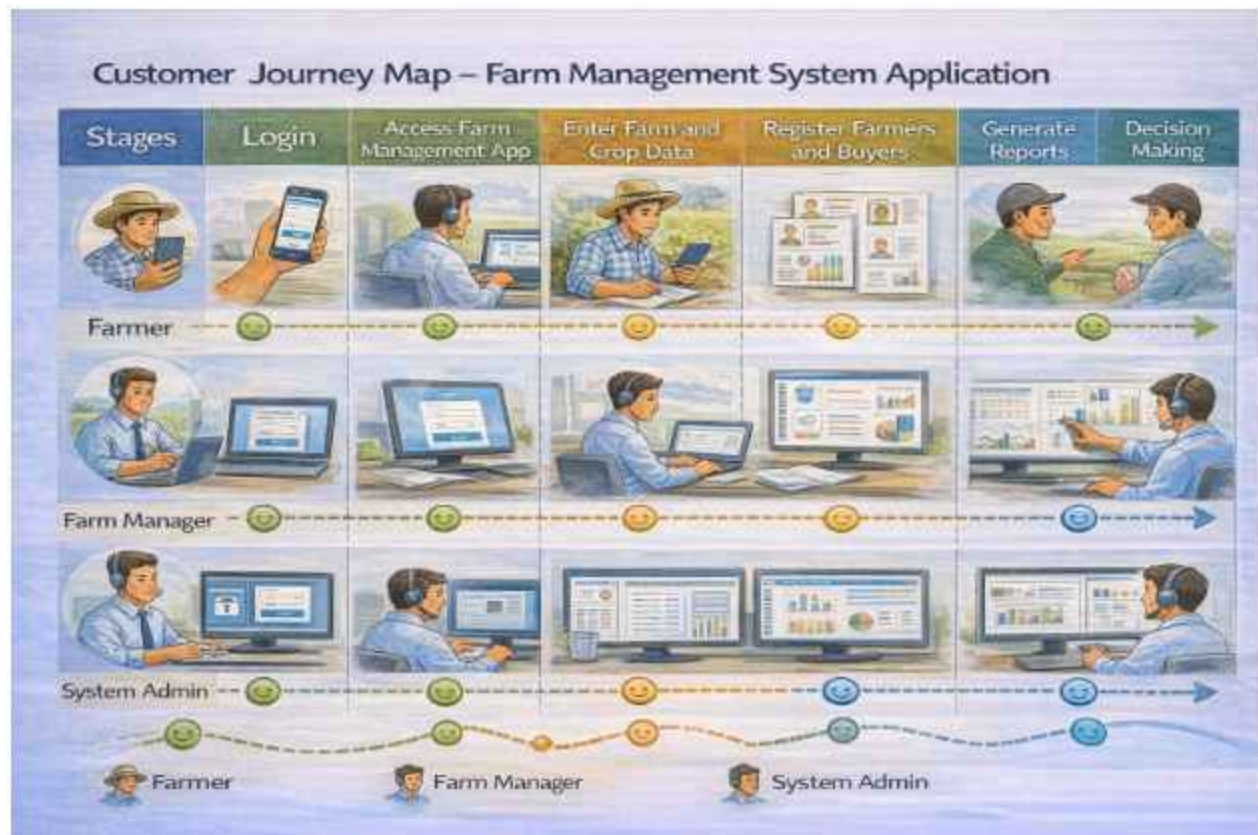


3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

1. Login
2. Access Farm Management App
3. Enter Farm and Crop Data
4. Register Farmers and Buyers
5. Generate Reports
6. Analyze Results
7. Decision Making



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	User Authentication
FR-2	Farm, Crop, Farmer, Buyer Management
FR-3	Data Entry and Validation
FR-4	Relationship Mapping
FR-5	Report Generation

FR-6	Dashboard Visualization
FR-7	User Role Management

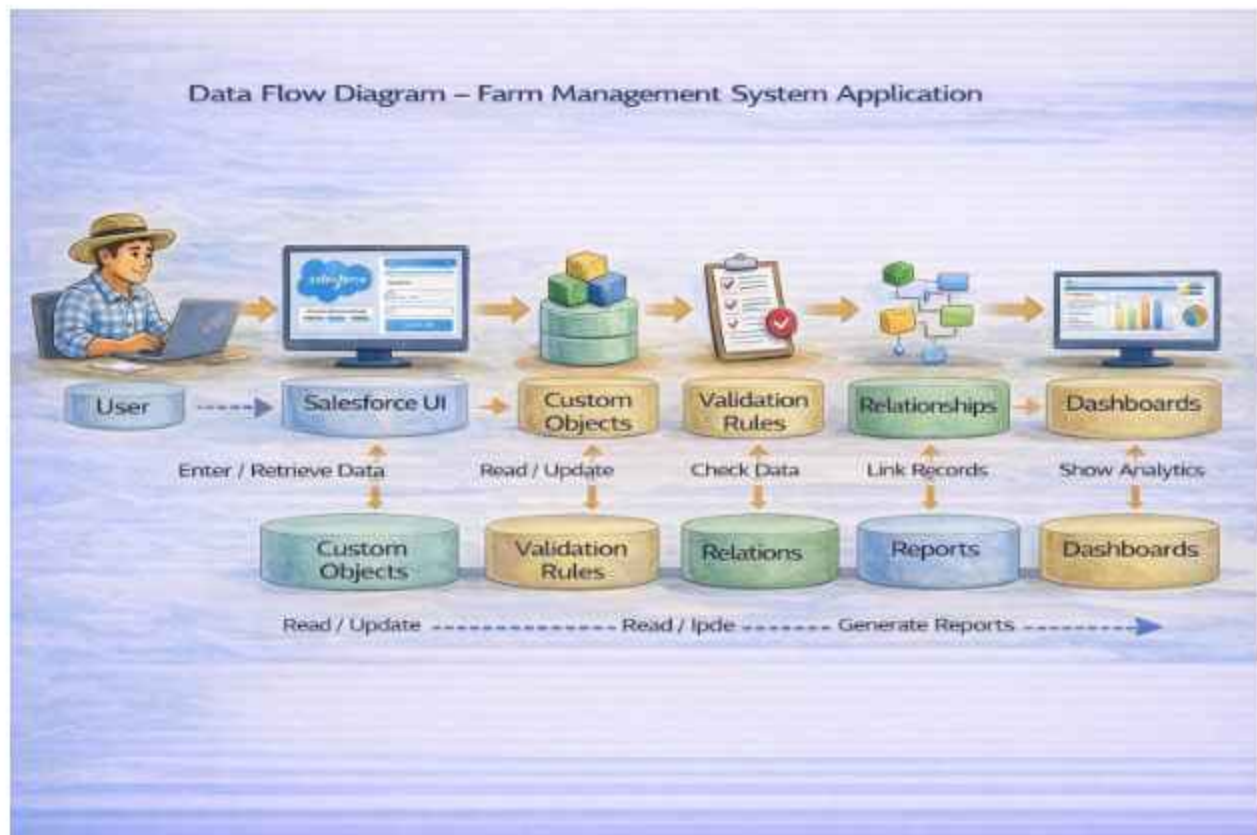
Non-Functional Requirements

NFR	Description	Example
Usability	Easy interface	Simple Lightning UI
Security	Data protection	Profiles & roles
Reliability	No data loss	Cloud backup
Performance	Fast access	Reports in <3 sec
Availability	24/7 access	Cloud hosting
Scalability	Handles growth	More farms/users

3.3 Data Flow Diagram

Data Flow:

User → Salesforce UI → Custom Objects (Farm, Crop, Farmer, Buyer) → Validation Rules → Relationships → Reports → Dashboards



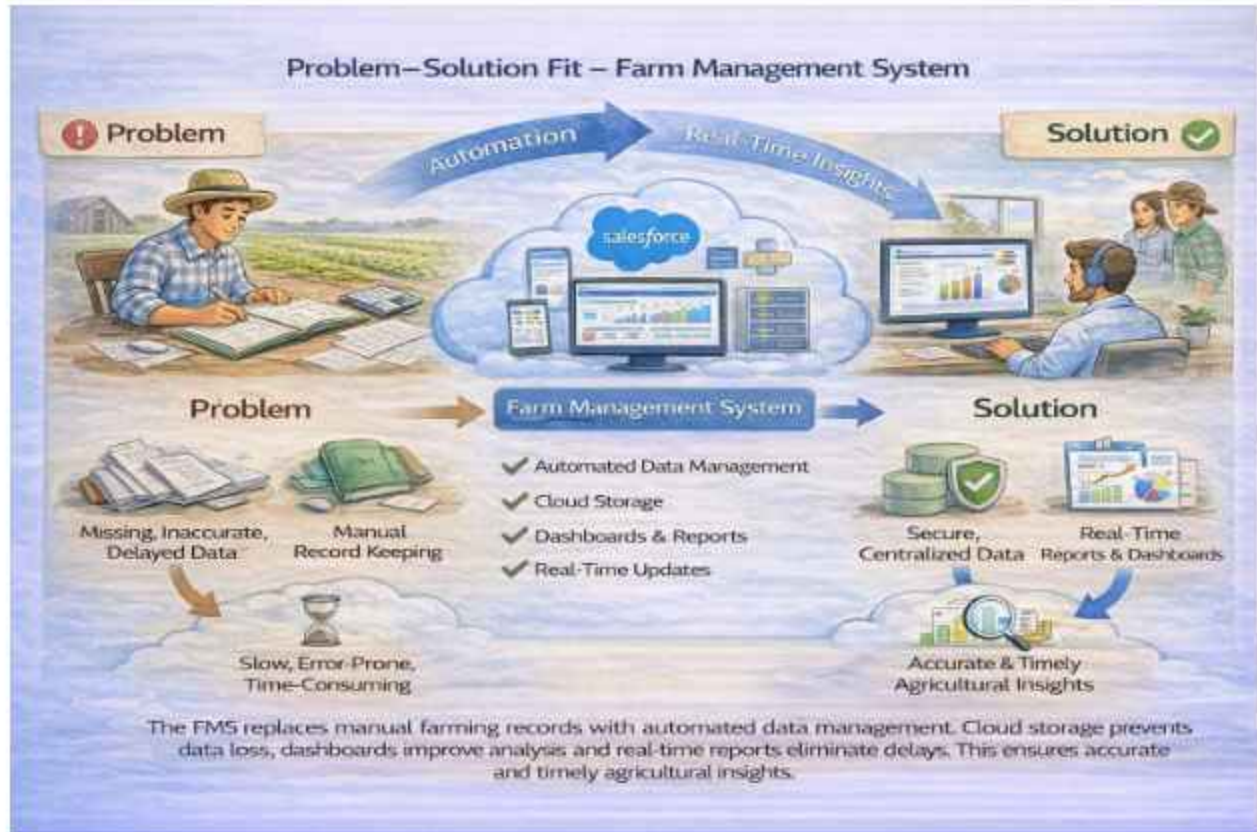
3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning UI
Backend	Salesforce Platform
Database	Salesforce Cloud
Automation	Validation Rules
Analytics	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The FMS replaces manual farming records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real-time reports eliminate delays. This ensures accurate and timely agricultural insights.



4.2 Proposed Solution

Functional Requirements

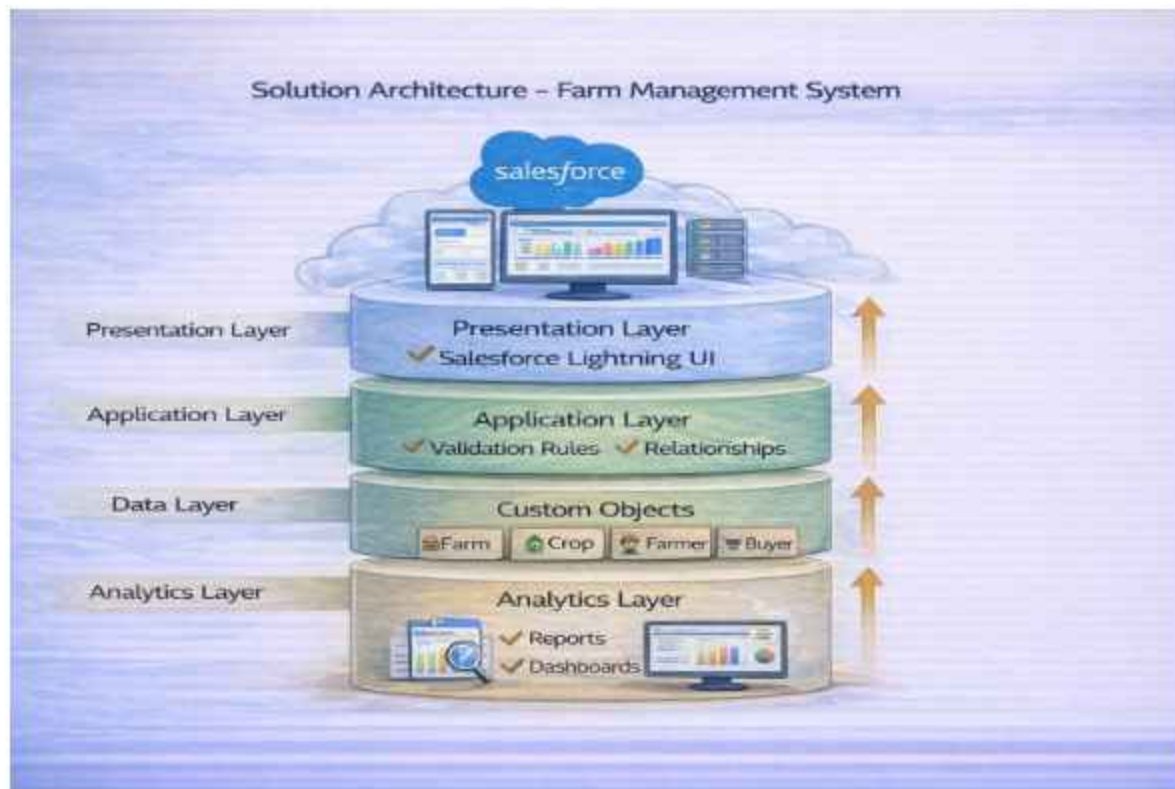
Parameter	Description
Problem	Manual maintenance of farm, crop, farmer, and buyer records leading to errors, data loss, and delays.

Solution	Salesforce-based Farm Management System with centralized data storage and automation.
Innovation	Automated data entry validation, cloud storage, and real-time reporting dashboards.
Impact	Improved data accuracy, faster reporting, and better decision-making for farmers and managers.
Business Model	Digital farm management and agricultural data services platform.
Scalability	Supports multiple farms, users, and regions with cloud-based infrastructure.

4.3 Solution Architecture

Architecture Layers

1. Presentation Layer – Salesforce Lightning UI
2. Application Layer – Validation Rules, Relationships
3. Data Layer – Custom Objects
4. Analytics Layer – Reports and Dashboards



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Farm Management System.	3	High	Member1
Sprint-2a	Data Modeling	USN-2	As an admin, I want to create Farm, Crop, Farmer, and Buyer objects with proper relationships so that agricultural data is stored and managed efficiently.	5	High	Member2
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all farm modules so that I can easily navigate and perform daily operations.	5	High	Member3
Sprint-3	Automation	USN-4	As a system, I want to validate farm and crop details such as dates, quantities, and mandatory fields so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a manager, I want automated flows for crop updates and buyer notifications so that farm operations are processed faster without manual intervention.	3	High	Member4
Sprint-4	Security	USN-6	As an administrator, I want role-based access control for farmers, managers, and buyers so that sensitive farm and transaction data is secure.	5	High	Member2
Sprint-5	Reports	USN-7	As a farm manager, I want reports showing crop yield, sales, and buyer transactions so that I can monitor farm performance and make informed decisions.	4	High	Member3
Sprint-6	Dashboards	USN-8	As a farm manager, I want dashboards showing crop productivity, sales trends, and operational statistics so that I can monitor performance visually.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

6. Project Development Phase:

Project Description:

The Farm Management System (FMS) in Salesforce is a comprehensive solution designed to digitize and streamline agricultural operations. This application empowers farmers, farm managers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

Milestone 1-Salesforce:

What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers.

Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity1:

Creating a Salesforce Developer Org:

A Developer org has all the features and licenses you need to get started with Salesforce.

1. Search [Developer.salesforce.com/signup](https://developer.salesforce.com/signup)
2. Enter the following details like First name, last name, Email, Role, Company, Country/Region, Postal code, and Username must be unique.
3. Click sign me up, after a few min you will receive a mail from salesforce org and by using the verify account link you can create your new password.
4. Click save.
5. Search login.salesforce.com

Build enterprise-quality apps fast to bring your ideas to life

Build apps fast with drag and drop tools
Customize your data model with clicks
Go further with Apex code
Integrate with anything using powerful APIs
Stay protected with enterprise-grade security
Customize UI with clicks or any leading-edge web framework

Sign up for your Salesforce Developer Edition
A full-featured copy of the Platform, for free

Complete the form to start your free trial. Our team will be in touch to help you make the most of your trial.

First Name*
Your first name

Last Name*
Your last name

Email*
Your email address

Role*
Your role

Company*
Company Name

Country/Region*
Country/Region

Postal Code*
Your postal code

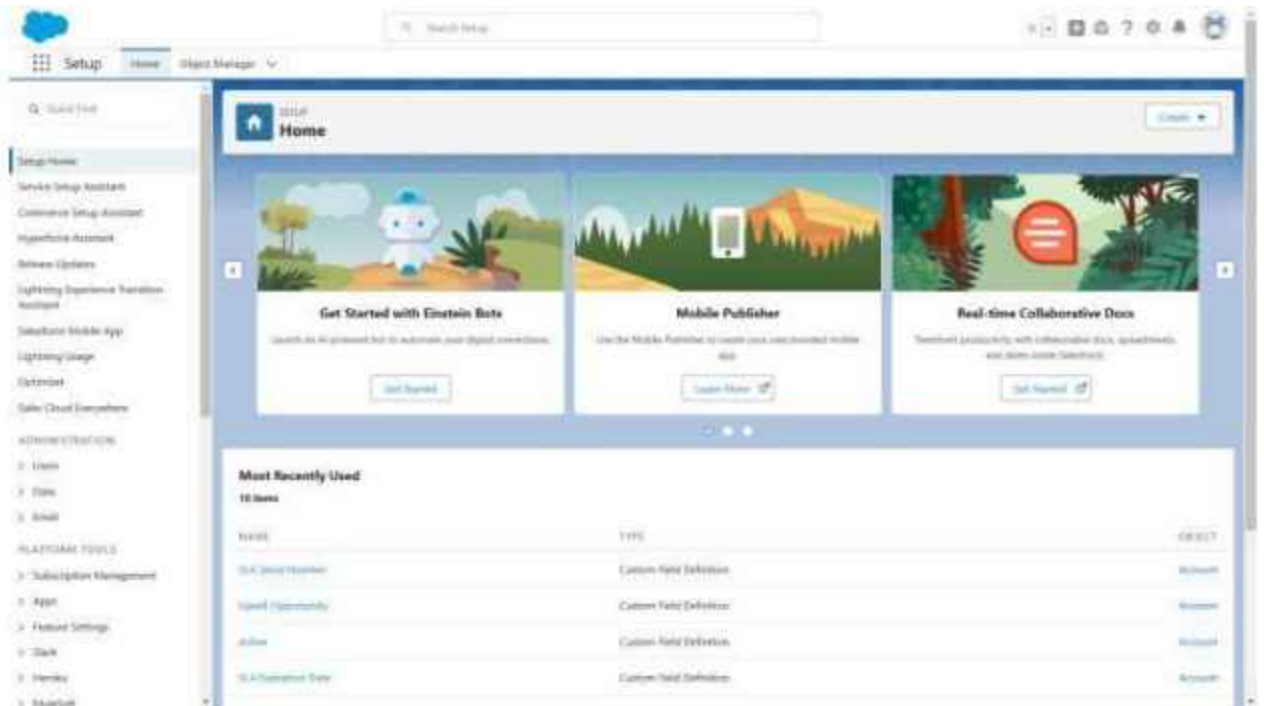
Username*
your@companyname.com

Your username should be in the form of your email address. It should have to be used in creating, editing and deleting the account with another Salesforce administrator. Don't worry about real-time recommendations.

☐ I agree to the Salesforce Developer Edition Terms of Service and Privacy Policy

Sign me up

6. By using username and password you can into the Salesforce org. The homepage will appear below



Milestone 2-Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

In This Application We Use 4 Custom Objects:

- 1.Farm
- 2.Crop
- 3.Farmer
- 4.Buyer

Activity 1:

Create a custom object for Farm:

To create a custom object, follow these steps:

1. From setup click on object manager.
2. Click create, select custom object.



3. Fill in the label as " Farm ".
4. Fill in the plural label as " Farms".
5. Record name: Farm Name"
6. Select the data type as "Text".

The screenshot shows the 'Create New Object' configuration page in SAP Fiori. The left sidebar contains a navigation menu with the following items: Fields & Relationships, Page Layouts, Highlighting Related Pages, Buttons, Links and Actions, Compact Layouts, Field Sets, Object Layouts, Related Pages, Related Lookup Pages, Search Layouts, List View Buttons Layout, Restriction Rules, Grouping Rules, and Object Access. The main content area is titled 'Data Type' and shows 'Text' selected. A warning message states: 'Warning: If you plan to store a high volume of records in this object, use the API for inserts. Use the Text field type.' The configuration is divided into several sections: 'Optional Features' with checkboxes for 'Allow Reports' (checked), 'Allow Archiving' (unchecked), 'Track Field History' (unchecked), 'Allow as Cluster Content' (unchecked), and 'Enable Locking' (unchecked); 'Object Classification' with checkboxes for 'Allow Mapping' (checked), 'Allow Bulk API Access' (checked), and 'Allow Streaming API Access' (checked); 'Deployment Status' with radio buttons for 'In Development' (unchecked) and 'Deployed' (checked), accompanied by a 'Show in App' link; and 'Search Status' with a checkbox for 'Allow Search' (checked). At the bottom, there are buttons for 'Save', 'Save & Close', and 'Cancel'.

7. In the Optional Features section, select Allow Reports and Track Field History.
8. In the Deployment Status section, ensure Deployed is selected.
9. In the Search Status section, select Allow Search.
10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
11. Leave everything else as is, and click Save.

Activity 2:

Creation of Crop object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Crop
 - c. Plural Label: Crops
 - d. Record Name: Crop Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox
 - g. Check the Allow Search checkbox
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout
 - i. Click Save.

Activity 3:

Creation of Farmer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Farmer

- c. Plural Label: Farmers
- d. Record Name: Farmer Name
- e. Select the data type as "Text".
- f. Check the Allow Reports checkbox.
- g. Check the Allow Search checkbox.
- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
- i. Click Save.

Activity 4:

Creation of Buyer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Buyer
 - c. Plural Label: Buyers
 - d. Record Name: Buyer Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox.
 - g. Check the Allow Search checkbox.
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
 - i. Click Save.

Milestone 3 -

Tab: Tab

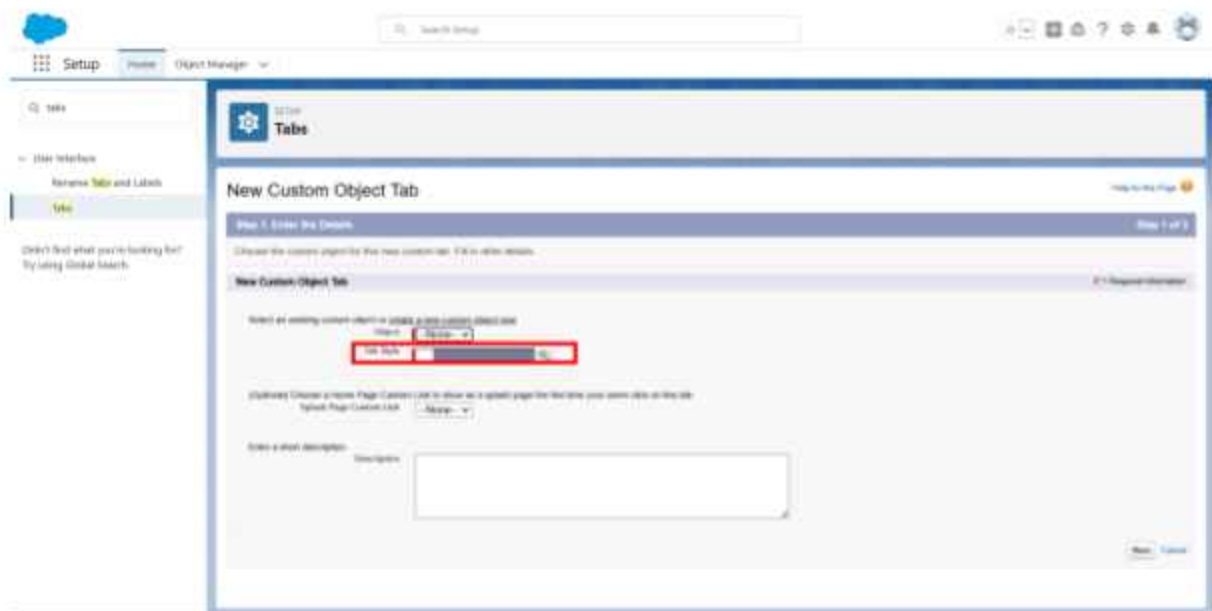
Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1:

Creation of Farm tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.



3. For Object, select Farm.

4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Crop.

4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farmer.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

Creation of Buyer tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Milestone 4 - Lightning App:

What is an App?

Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

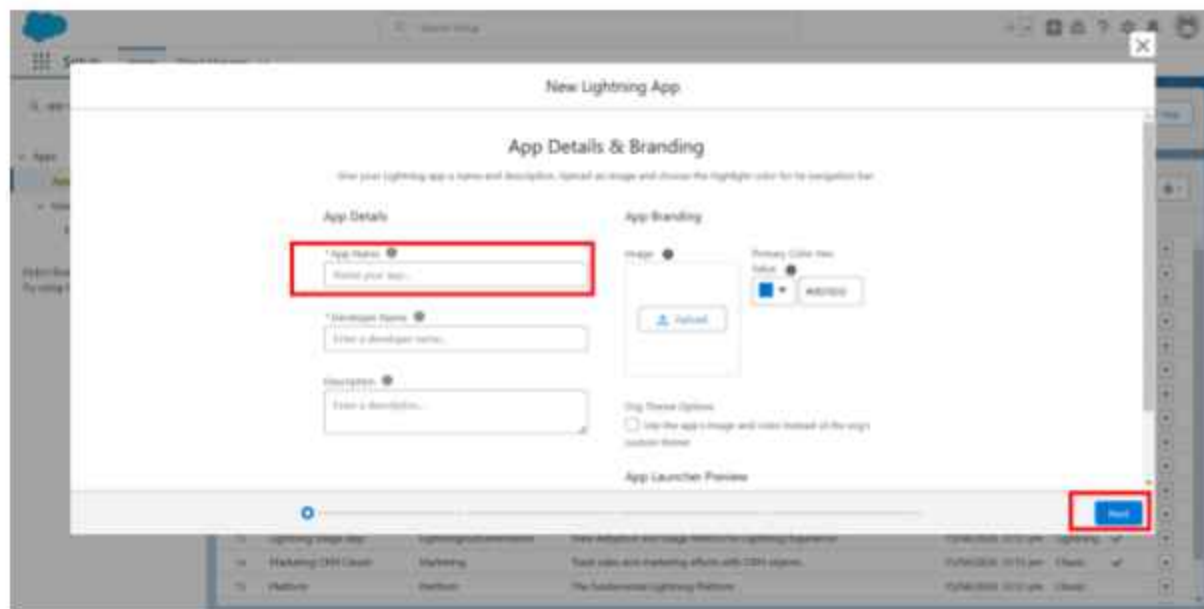
Activity 1:

Create the Farm Management System app

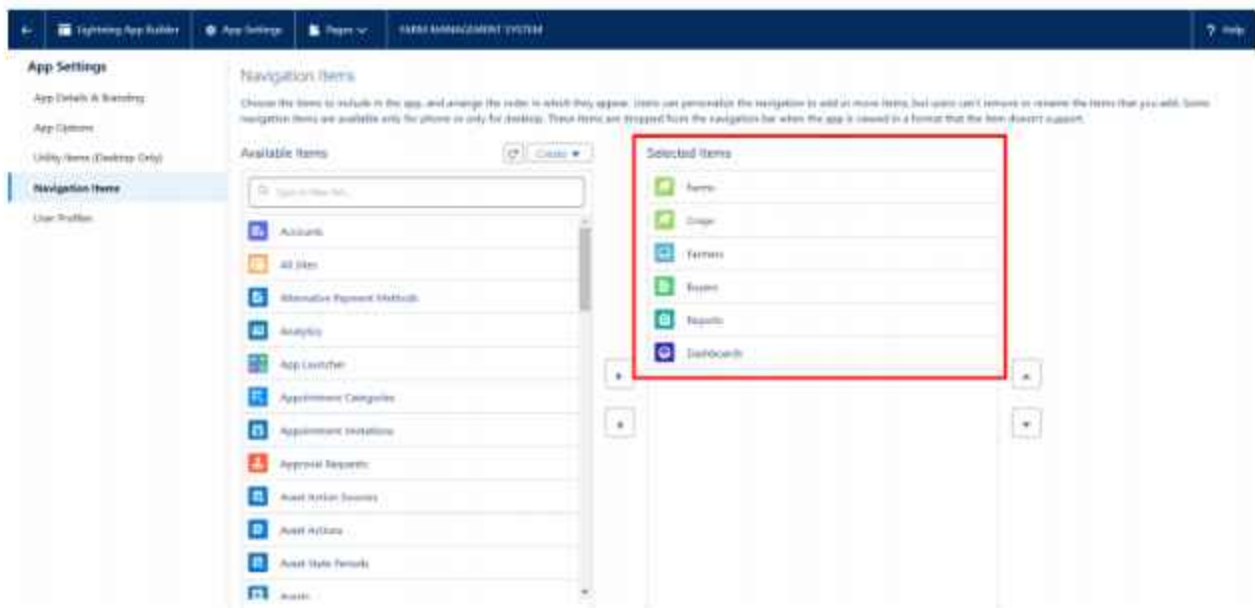
The screenshot shows the Salesforce Lightning Experience App Manager interface. The left sidebar contains navigation links: Setup, Home, Object Manager, App Manager, External Sites App, and External Client App Manager. The main content area displays a list of apps with columns for App Name, Developer Name, Description, Last Modified, App Type, and Status. The list includes apps like AI Tools, Analytics Studio, App Launcher, Automation, Bulk Solutions, Business Rules Engine, Community, Content, Data Manager, Digital Experience, Email Marketing, Lightning Stage App, Marketing CRM Cloud, and Platform.

App Name	Developer Name	Description	Last Modified	App Type	Status
1 AI Tools	AI Tools	AI Tools	10/10/2024, 10:10 am	Classic	✓
2 Analytics Studio	Analytics	Built for Analytics Studio with an app	10/10/2024, 10:10 am	Classic	✓
3 App Launcher	App Launcher	App Launcher tabs	10/10/2024, 10:10 am	Classic	✓
4 Automation	Automation	Automated business processes and repetitive tasks	10/10/2024, 10:10 am	Lightning	✓
5 Bulk Solutions	LightningBulk	Streamline and manage business solutions designed for your industry	10/10/2024, 10:10 am	Lightning	✓
6 Business Rules Engine	BusinessRulesEngine	Create and manage business rules that perform complex logic and calculations	10/10/2024, 10:10 am	Lightning	✓
7 Community	Community	Salesforce CRM Community	10/10/2024, 10:10 am	Classic	✓
8 Content	Content	Salesforce CRM Content	10/10/2024, 10:10 am	Classic	✓
9 Data Manager	DataManager	Use Data Manager to view data, monitor usage, and manage usage	10/10/2024, 10:10 am	Lightning	✓
10 Digital Experience	DigitalExperience	Manage content and media for all of your sites	10/10/2024, 10:10 am	Lightning	✓
11 Email Marketing	EmailMarketing	Email Marketing	10/10/2024, 10:10 am	Lightning	✓
12 External Client App Manager	ExternalClientAppManager	External Client App Manager	10/10/2024, 10:10 am	Lightning	✓
13 Lightning Stage App	LightningStageApp	View, deploy, and manage Lightning Experience	10/10/2024, 10:10 am	Lightning	✓
14 Marketing CRM Cloud	Marketing	Track sales and marketing efforts with CRM system	10/10/2024, 10:10 am	Classic	✓
15 Platform	Platform	The foundation of Lightning Platform	10/10/2024, 10:10 am	Classic	✓

- From Setup, enter App Manager in the Quick Find and select App Manager.
 - Click New Lightning App. Farm Management System as the App Name, then click Next
 - Under App Options, leave the default selections and click Next.
 - Under Utility Items, leave as is and click Next.
 - From Available Items, select Farm, Crop, Farmer, Buyer Reports, and Dashboards and move them to Selected Items. Click Next. • From Available Profiles, select System Administrator and move it to Selected Profiles. Click Save & Finish.
 - To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.
- Note:
1. App Launcher-Displays available apps.
 2. App Name-Displays the current selected app.



3. Navigation menu-Displays the tabs available inside the app.



Milestone 5 -

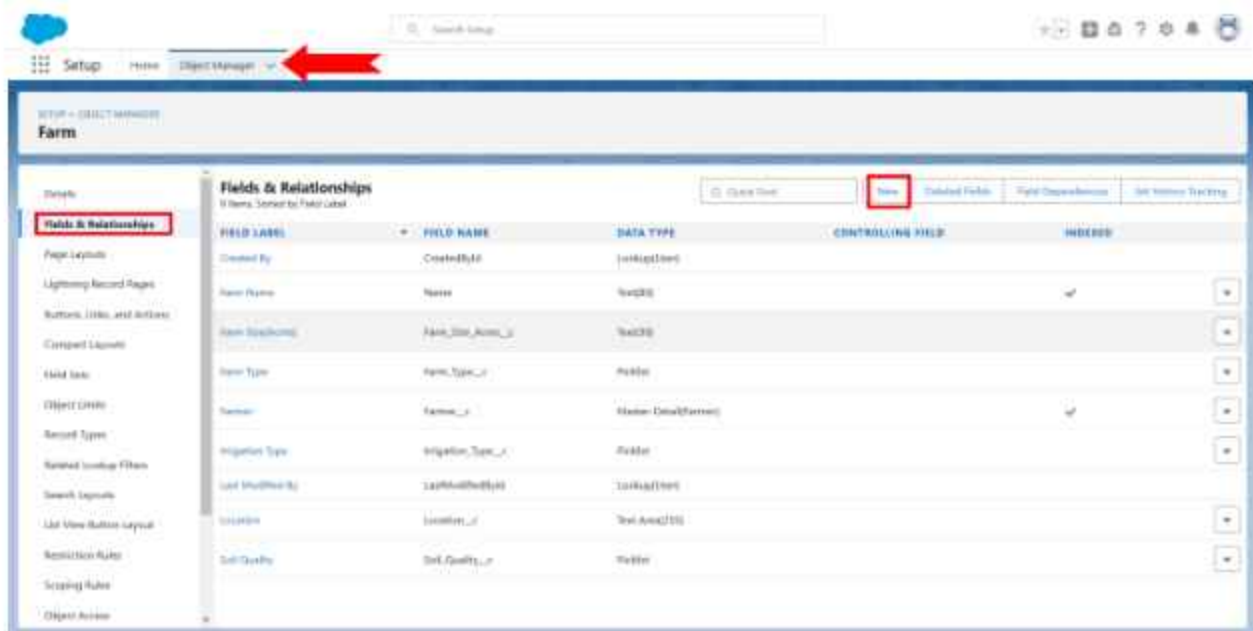
Fields and Relationship: What are fields?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

Activity 1:

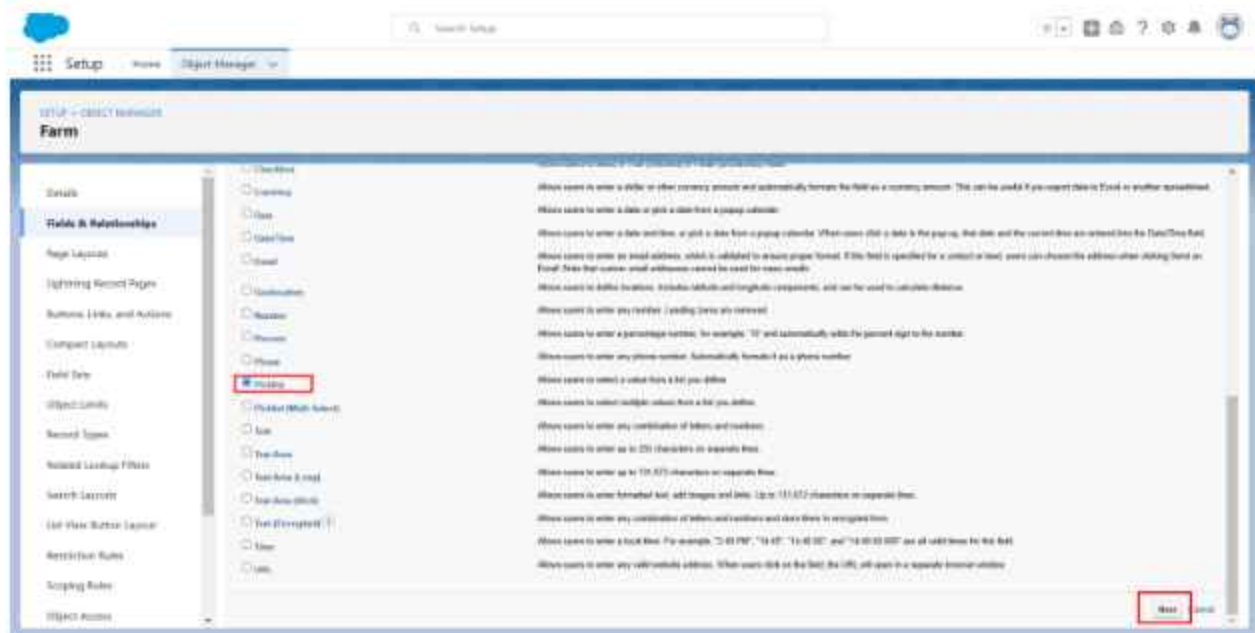
Creation of fields for the Farm objects:

1. click the gear icon and select Setup. This launches Setup in a new tab.
2. Click the Object Manager tab next to Home.
3. Select Farm.
4. Select Fields & Relationships from the left navigation, and click New



FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedBy	Lookup (User)		
Name	Name	Text		✓
Name (Duplicate)	Name_Duplicate	Text		
Name Type	Name_Type	Text		
Name	Name	Master-Detail (Farm)		✓
Irrigation Type	Irrigation_Type	Text		
Leaf (Number)	Leaf (Number)	Lookup (Leaf)		
Location	Location	Text Area (10)		
Leaf Quality	Leaf_Quality	Text		

5. Choose Data type as a picklist, Click next



6. Enter Field Label (Farm Type), and Enter Values as (Crop, Livestock, Mixed)
7. Next, Next Click Save.

Setup > OBJECT MANAGER

Farm

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links and Actions

Component Libraries

Help Gate

Object Links

Record Types

Related Lookup Fields

Search Layouts

User View Button Layout

Validation Rules

Scoping Rules

Object Access

New Custom Field

Step 1: Enter the Details

Step 1 of 4

Field Label: Farm Type

Values:

☐ Use global picklist value set

☒ Enter values with each value separated by a new line

Crop
Livestock
Mixed

☐ Display values alphabetically, not in the order entered

☐ Use first value as default value

☒ Manual control in the values defined in the value set

Field Name: Farm_Type

Description:

Help Text:

Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

- 1.Next we Create another Picklist Field,Enter field label (Soil Quality) enter values (Organic,Cover Crops,Composting)
- 2.Again we Create Picklist field,Field Label (Irrigation type), Enter Values (Sprinkler irrigation,Drip irrigation,localized irrigation)
- 3.Choose data type as a text,enter field label(Location) 4.Choose data type as text,enter field label (size)
- 5.Create a Master Detail relationship with Farmer Object.

Activity 2:

Creation of fields for the Crop objects:

- 1.Choose Data type As a text,Enter Field Label name (Crop Name),Next save. 2.Select Data type as date ,Enter Field label (Planting Date),then save .
- 3.Again we select data type as a date,Enter field labe (Harvest date),save
- 4.Next We Create Lookup relationship on crop object realted to Farm Object.
- 5.We select adata type as currency,Enter field Label (Actual Yield), Next Save it.
- 6.Select Data type as a Number,enter Field Label (Total Acres Harvested),Enter Maximum Length(20) and save.
- 7.Now We create Formula Field For That Select data Type As a Formula,Click Next.



Crop

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Links

Record Types

Related Lookup Filters

Search Layouts

Set View Button Layout

Assignment Rules

Sharing Rules

Object Access

Specify the type of information that the system field will contain

Data Type

☐ Auto-Increment

Select one of the data types below

☐ Auto Number

A system-generated sequential number that uses a display format you define. The number is automatically incremented for each new record.

☒ Currency

A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the values it relies on change.

☐ Date/Time

A read-only field that displays the year, month, or median value of a field in a related list or the record count of all records listed in a related list.

☐ Lookup Relationship

Creates a relationship that links from object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.

☐ Master-Detail Relationship

Creates a special type of parent-child relationship between two objects (the child or "lookup" and master objects (the parent or "master") values.

- The relationship field's resolution of all detail records.
- The relationship and sharing of detail records are determined by the master record.
- When a new parent for master record, all detail records are deleted.
- The user creates other summary fields on the master record to summarize the detail records.

The relationship field allows users to click on a lookup icon to select a value from a popup list. The master object is the source of the values in the list.

Creates a relationship that links from object to an external object whose data is stored outside the Salesforce org.

☐ Parent-Child Relationship

When users to select a field (parent), it value (parent) value.

☐ Picklist

When users to enter a value in other summary, select and automatically through the field as a summary amount. This can be used to pre-select data in fields or another spreadsheet.

☐ Summary

When users to enter a value in pick a value from a popup window.

☐ Text

When users to enter a date and time, or pick a date from a popup window. When users that a date in the popup, that date will be current time as entered into the Date/Time field.

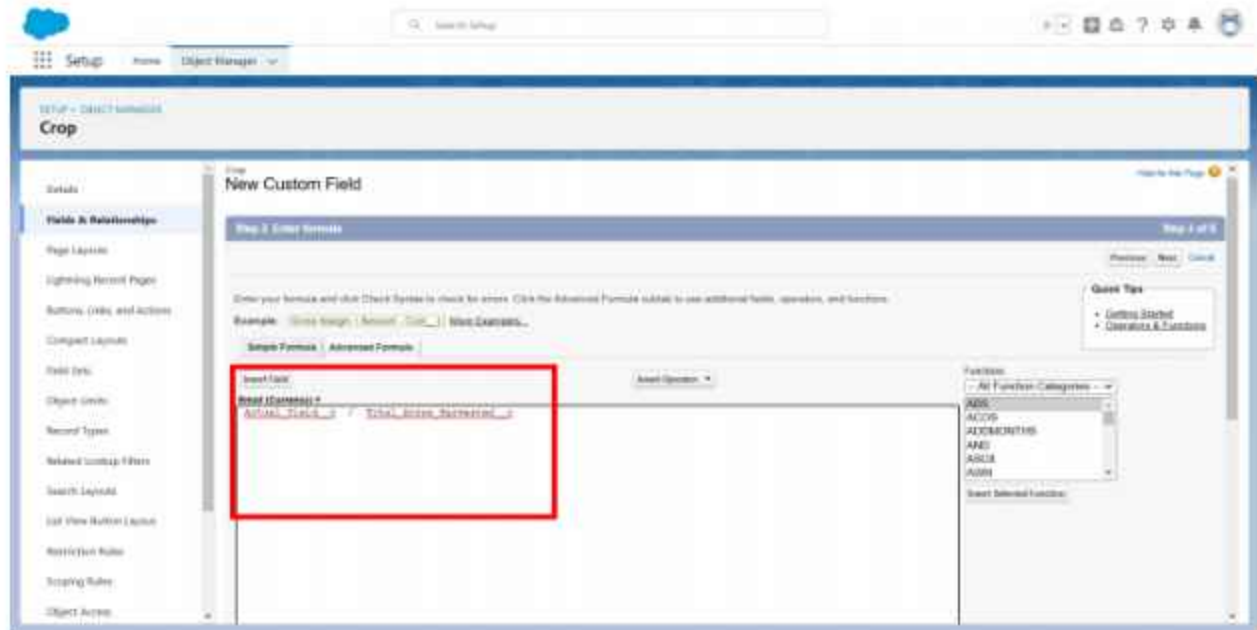
☐ Text Area

When users to enter a record address, which is different to ensure proper format. If the field is specified in a related list, users can choose the address when clicking "Add or Edit". Note that address and address cannot be used for text area.

☐ Text (Rich)

8. Enter Field Label (Expected Yield per Acre), Select formula return type as Currency, click Next.

9. Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.



Activity 3:

Creation of fields for the Farmer objects:

1. Select Data type Phone, Enter field label(Contact Number), next Save.
2. Select Data type Picklist, Enter Field Label(State), Enter values (Telangana, Andhra Pradesh, Tamilnadu), save.
3. Choose Data type Picklist, Enter Field Label (District), Enter Values of district, next save.
4. Again select Data type as Picklist, Enter Field Label (Village), Enter Values of villages, next save.

Creation of field dependency :

1. Now create Field Dependency For District field, Open Object Manager, Choose Farmer object, select Fields&Relationships, On Right side We have Field Dependency option, select that option.
2. Click new, Select Controlling Field as State, Dependent Field As District, Click on Continue.
3. Select the appropriate Value in each column by double-clicking them.
4. For Ex. Telangana - Yadadri Bhongir.
5. Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.
6. Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.



Search Setup



Setup > Object Manager > **Fields**

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Labels

Record Types

Related Lookup Fields

Search Layouts

List View Button Layout

Restriction Rules

Scoping Rules

Object Access

New Field Dependency

Create a dependent relationship that causes the values of a picklist or multi-select field to be determined based on the value selected in the user or another field.

- The field that drives filtering is called the "controlling field."
- Standard and custom picklists and picklists with global pick and roll lists can be controlling fields.
- The field that has its values filtered is called the "dependent field."
- Custom picklists and multi-select picklists can be dependent fields.

Step 1: Select a controlling field and a dependent field. (Click Continue when finished.)

Step 2: On the following page, add the filter rules that control the values that appear in the dependent field for each value in the controlling field.

Controlling Field: **Picklist**

Dependent Field: **Picklist**

Continue Cancel

Continue Cancel

Setup

Home

Object Manager

Search Setup

Quick Start

Setup Home

Service Setup Assistant

Commerce Setup Assistant

Hyperforce Assistant

Release Updates

Lightning Experience Transition Assistant

Salesforce Mobile App

Lightning Design System

Optimizer

Sales Cloud Everywhere

ADMINISTRATOR

Users

Data

Email

PLATFORM TOOLS

Subscription Management

Apps

System Settings

Setup

Dependency Field

Details

Instructions

- Click on the on or off to toggle its visibility for the Controlling Field value shown in the column heading.
- To change multiple rows at once, select multiple rows and then click the Include Values or Exclude Values button to change the visibility of all selected rows at once.
- Use SHIFT + click to select a range of adjacent rows. Use CTRL + click to select multiple rows that are not adjacent.
- Use the Filter button to find the results.

Legend

Excluded Value

Included Value

Click button to include or exclude selected rows from the dependent column.

Include Values

Exclude Values

Row	Dependency	Dependency Field	Dependency Value
1	Account	Account Name	Account Name
2	Account	Account Name	Account Name
3	Account	Account Name	Account Name
4	Account	Account Name	Account Name
5	Account	Account Name	Account Name
6	Account	Account Name	Account Name
7	Account	Account Name	Account Name
8	Account	Account Name	Account Name
9	Account	Account Name	Account Name
10	Account	Account Name	Account Name
11	Account	Account Name	Account Name
12	Account	Account Name	Account Name
13	Account	Account Name	Account Name
14	Account	Account Name	Account Name
15	Account	Account Name	Account Name
16	Account	Account Name	Account Name
17	Account	Account Name	Account Name
18	Account	Account Name	Account Name
19	Account	Account Name	Account Name
20	Account	Account Name	Account Name
21	Account	Account Name	Account Name
22	Account	Account Name	Account Name
23	Account	Account Name	Account Name
24	Account	Account Name	Account Name
25	Account	Account Name	Account Name
26	Account	Account Name	Account Name
27	Account	Account Name	Account Name
28	Account	Account Name	Account Name
29	Account	Account Name	Account Name
30	Account	Account Name	Account Name
31	Account	Account Name	Account Name
32	Account	Account Name	Account Name
33	Account	Account Name	Account Name
34	Account	Account Name	Account Name
35	Account	Account Name	Account Name
36	Account	Account Name	Account Name
37	Account	Account Name	Account Name
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58	Account	Account Name	Account Name
59	Account	Account Name	Account Name
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91	Account	Account Name	Account Name
92	Account	Account Name	Account Name
93	Account	Account Name	Account Name
94	Account	Account Name	Account Name
95	Account	Account Name	Account Name
96	Account	Account Name	Account Name
97	Account	Account Name	Account Name
98	Account	Account Name	Account Name
99	Account	Account Name	Account Name
100	Account	Account Name	Account Name

Click button to include or exclude selected rows from the dependent column.

Include Values

Exclude Values

Done

Cancel

Refresh

7. Click Close to close the preview window, Click Save.

Activity 4:

Creation of fields for the Buyer objects:

1. Select Data type as a Phone, Enter a Field Label(Contact), Save.
2. Select Data type as an Email, Enter Field Label(Email), Next save.
3. Select Data type Text area, Enter Field Label(Address), Save it.
4. Select Data type Picklist, Field Label(Buyer Type), Enter values (Wholesale, Retail), Click Save.

Milestone 6- Profile:

What is a profile?

A profile is a group/collection of settings and permissions that define what a user can do in

salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab

settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.

A profile can be assigned to many users, but user can be assigned single profile at a time.

Activity 1:

Create a custom profile:

1. From setup , enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, enter Sales Department.



Search Setup

Setup | Overview | Object Manager

profile

Overview | **Profiles**

Didn't find what you're looking for?
Try using Global Search.

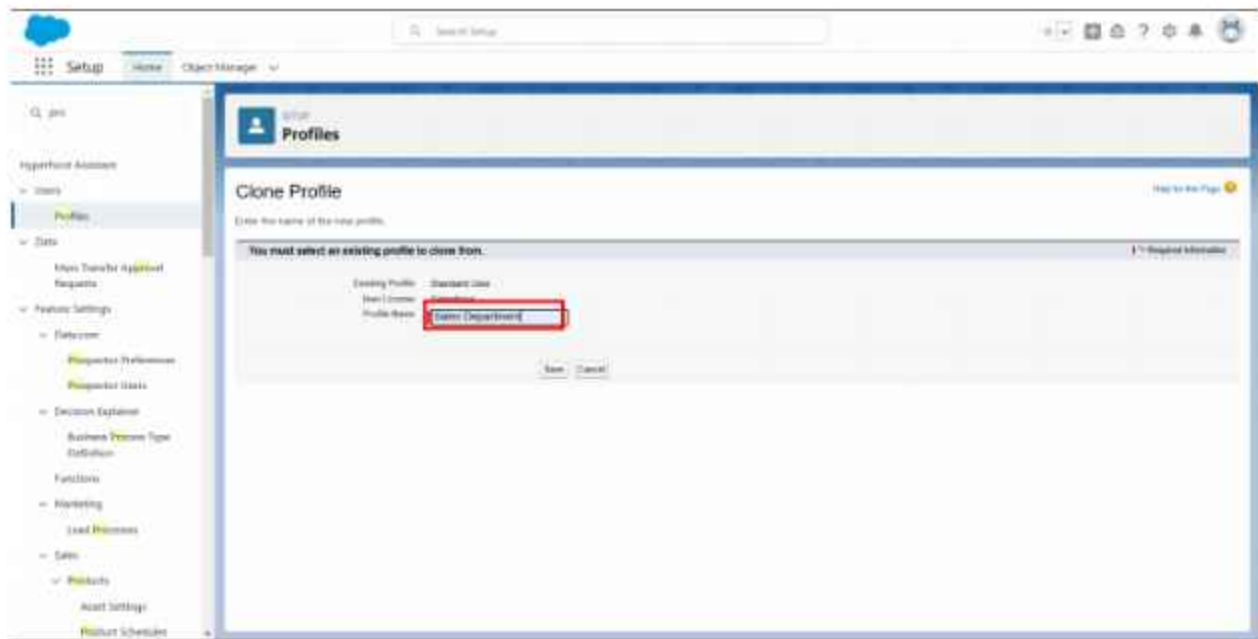
Profiles

Help for this Page

All Profiles

New Profile

Action	Profile Name	Start Location	Options
☐	Prod (Dev) - Data Development	Subsource Platform	☒
☐	Prod (Dev) - Data Analytics	Subsource	☒
☐	Prod (Dev) - Data Warehouse	Subsource Integration	☒
☐	Prod (Dev) - Data Platform (Dev)	Subsource	☐
☐	Prod (Dev) - Data Platform (Prod)	Subsource Platform	☐
☐	Prod (Dev) - Data Platform (Test)	Subsource Platform	☐
☒	Prod (Dev) - Standard Data	Subsource	☒
☐	Prod (Dev) - Subsource Administration	Subsource	☐



5. Click save.

Activity 2:

Create a custom profile-2:

Create a profile with the profile name as “ Farm Manager” .

6. From setup , enter profiles in Quick Find box
7. Select profiles (Standard user).
8. Click clone.

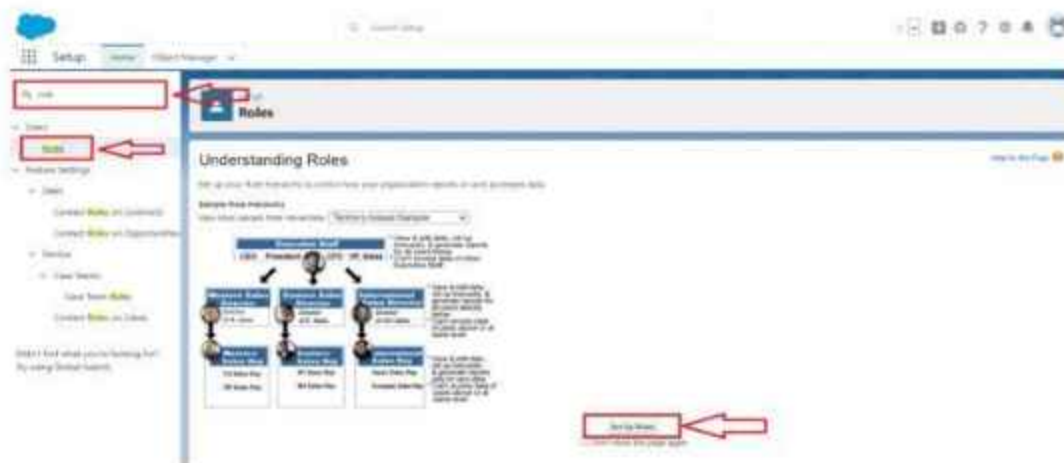
Milestone 7- Role

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

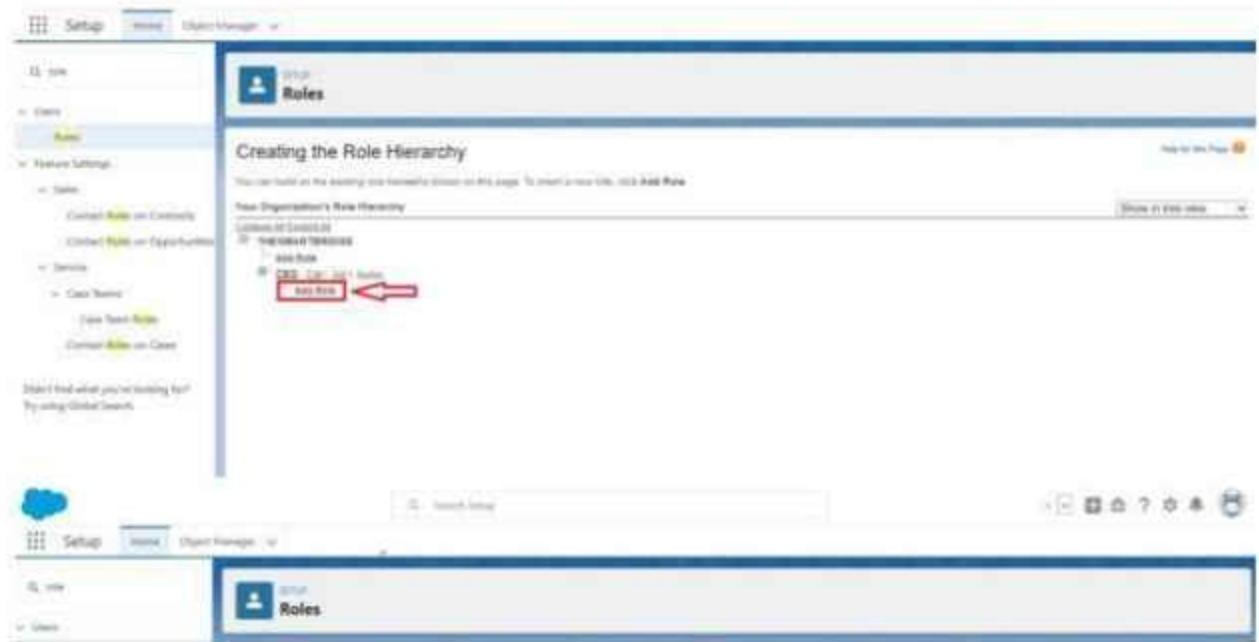
Activity-1

Creation of role

1. From the Quick find box search for the role and click on the roles option
2. select the set up roles option



3. Below the CEO click on add role and enter the label name as a " Farm Manager " and role name will be Automatically populated and click on save.



Milestone 8-User :

What is a user?

A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in

Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, so If you don't find salesforce license then deactivate a user who has salesforce license Or change the license type from Salesforce to any other.

Activity 1:

To Create a user:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.

The screenshot shows the Salesforce Setup interface. In the left sidebar, the 'Users' link under 'Manage Users' is highlighted with a red box. The main content area is titled 'All Users' and shows a table of existing users. A red box highlights the 'New User' button at the top of the table. The table has columns for 'Active', 'Full Name', 'Email', 'Username', 'Role', 'Active?', and 'Profile'. The 'Email' column contains email addresses, and the 'Username' column contains usernames. The 'Active?' column has checkboxes, and the 'Profile' column lists the assigned profile for each user.

Active	Full Name	Email	Username	Role	Active?	Profile
☐	Developer1	dev1@domain.com	dev1@domain.com	Developer	☒	Developer Profile
☐	Developer2	dev2@domain.com	dev2@domain.com	Developer	☒	Developer Profile
☐	Developer3	dev3@domain.com	dev3@domain.com	Developer	☒	Developer Profile
☐	Developer4	dev4@domain.com	dev4@domain.com	Developer	☒	Developer Profile
☐	Developer5	dev5@domain.com	dev5@domain.com	Developer	☒	Developer Profile
☐	Developer6	dev6@domain.com	dev6@domain.com	Developer	☒	Developer Profile
☐	Developer7	dev7@domain.com	dev7@domain.com	Developer	☒	Developer Profile
☐	Developer8	dev8@domain.com	dev8@domain.com	Developer	☒	Developer Profile
☐	Developer9	dev9@domain.com	dev9@domain.com	Developer	☒	Developer Profile
☐	Developer10	dev10@domain.com	dev10@domain.com	Developer	☒	Developer Profile
☐	Developer11	dev11@domain.com	dev11@domain.com	Developer	☒	Developer Profile
☐	Developer12	dev12@domain.com	dev12@domain.com	Developer	☒	Developer Profile
☐	Developer13	dev13@domain.com	dev13@domain.com	Developer	☒	Developer Profile
☐	Developer14	dev14@domain.com	dev14@domain.com	Developer	☒	Developer Profile
☐	Developer15	dev15@domain.com	dev15@domain.com	Developer	☒	Developer Profile
☐	Developer16	dev16@domain.com	dev16@domain.com	Developer	☒	Developer Profile
☐	Developer17	dev17@domain.com	dev17@domain.com	Developer	☒	Developer Profile
☐	Developer18	dev18@domain.com	dev18@domain.com	Developer	☒	Developer Profile
☐	Developer19	dev19@domain.com	dev19@domain.com	Developer	☒	Developer Profile
☐	Developer20	dev20@domain.com	dev20@domain.com	Developer	☒	Developer Profile

3. Enter First name as Farm and last name as Manager.
4. Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.
5. Select the role Farm Manager.
6. Select user License as Standard Platform User.
7. Select profile(Farm Manager).
8. Click save.

Activity 2:

Create a user with a username as "Sales Department", and assign him the sales Manager profile. Follow the steps from above Activity

Milestone 9-Sharing Rules:

What are Sharing Rules?

Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.

Types of sharing rules,

- Owner-based Sharing Rules
- Criteria-based Sharing Rules

Activity 1:

Create a sharing rule

- 1) Goto Sharing Settings, which can be found under the Quick Find section.
- 2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.
- 3) Add the label of the sharing rule you want to make.
- 4) Select your rule type based on the criteria.
- 5) Select the field can join immediately check field from the Crop object.
- 6) Select the Crop as equal and value is True.
- 7) And in selecting the users to share with the section select roles and in that select Farm Manager.
- 8) And in the section of select the level of access for the users give the access Read/Write.
- 9) And save the rule.

Setup

Share

Client Manager

Sharing

Security

Sharing Settings

Guest User Sharing Rule Setup Report

Sharing Settings

Default list what you're looking for by using Global Search

Search Settings

Sharing Settings

Use sharing rules to make automatic decisions to your organization-wide sharing settings for defined sets of users.
Note: "Roles and subordinates" includes all users in a role, and the roles below that role.
You can use sharing rules only to grant roles to roles, not to restrict access.

Label:

Rule Name:

Description:

Step 1: Select your rule type

Fields

Field	Operation	Value	AND
Corp Name	equals		AND
Name	equals		AND
Name	equals		AND
Name	equals		AND
Name	equals		AND

NOTE: This is a

☐ include records owned by users who are listed as assigned role

☐ exclude records owned by users who are listed as assigned role

Share with:

Access Level:

Created By:

Modified By:

Save

Cancel

Activity 2:

Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.

Follow the steps from above Activity.

Milestone-10: User Adoption

Activity 1: Create a Record(Farmer)

- Click on App Launcher on left side of screen.
- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.

The screenshot shows a web application interface for 'FARM MANAGEMENT'. A modal window titled 'Edit Karsava' is open, displaying a form for creating or editing a farmer record. The form includes the following fields:

- Farmer Name:** A text input field containing the name 'Karsava'.
- Contact Number:** A text input field containing the number '984262779'.
- Date:** A dropdown menu showing 'Add Date Placeholder'.
- Gender:** A dropdown menu showing 'Chillara'.
- Village:** A dropdown menu showing 'Nandella'.
- Land Size (Acres):** A text input field containing the value '15'.

At the bottom of the form, there are three buttons: 'Cancel', 'Save & Close', and 'Save'. The 'Save' button is highlighted with a red rectangular box.

Milestone 11-Reports:

What are Reports?

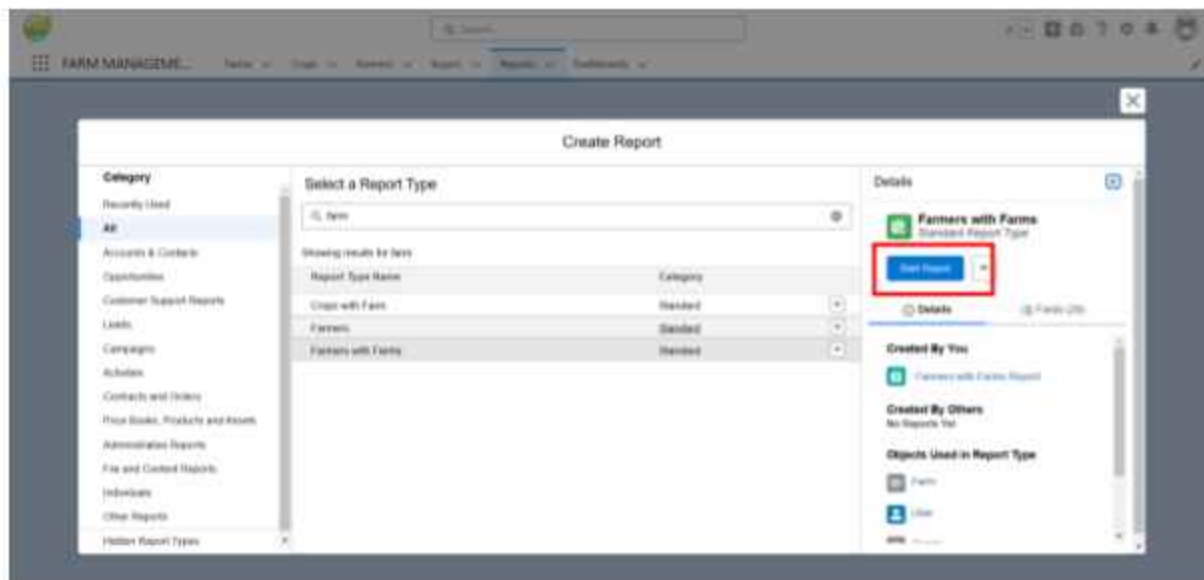
A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is stored in a folder. Folders can be public, hidden, or shared, and can be set to read-only or read/write.

Activity 1:

Create a report:

Create a report that displays rating of the account and which has type and account name.

- 1) Click on app launcher search for reports.
- 2) Click on the new report and select the category has Farmer with Farms.



- 3) In the details section select the option start report.
- 4) In Outline Click on Column Select Which Field you want to display in the Report.
- 5) Click Group,Select Which Fields You Want to Group.
- 6) Save the report by giving label name(Farmer with Farms) and save the folder as a public folder and save the report.

The screenshot displays the 'FARM MANAGEMENT' software interface. The top navigation bar includes 'Farms', 'Crops', 'Farmers', 'Reports', and 'Dashboard'. The 'Reports' section is active, showing a report titled 'Farmer with Farms Report' with a sub-label 'Farmer with Farms'. The report is displayed in a table format with the following columns: 'Irrigation Type', 'Farmer/Farmer Name', 'Farm/Farm Name', 'Cotton Number', 'Farm Size/area', and 'Farm Type'. The table contains three data rows and a total row. The 'Columns' section on the left shows the selected fields for the report: 'Farmer/Farmer Name', 'Farm/Farm Name', 'Cotton Number', 'Farm Size/area', and 'Farm Type'. The 'Groups' section on the left shows the selected fields for grouping: 'Irrigation Type' and 'Farm Type'. The 'Status' section at the bottom shows the report is 'Completed'.

Irrigation Type	Farmer/Farmer Name	Farm/Farm Name	Cotton Number	Farm Size/area	Farm Type
Regain	Paddy Farm	100000000	10	1000000	10
Regain	Regain Farm	100000000	10	1000000	10
Regain	Regain Farm	100000000	10	1000000	10
Total				30	

Activity 2:

Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

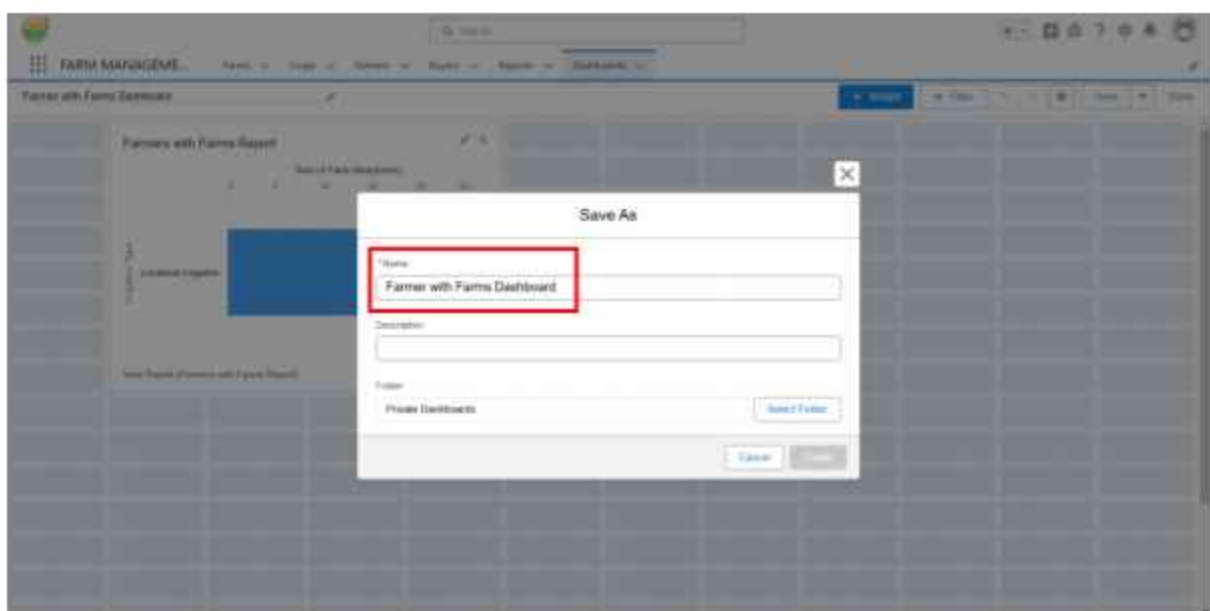
Milestone 12-Dashboard:

Dashboards provide more insights than reports as they combine the data from many reports and show a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity 1:

Create a Dashboard:

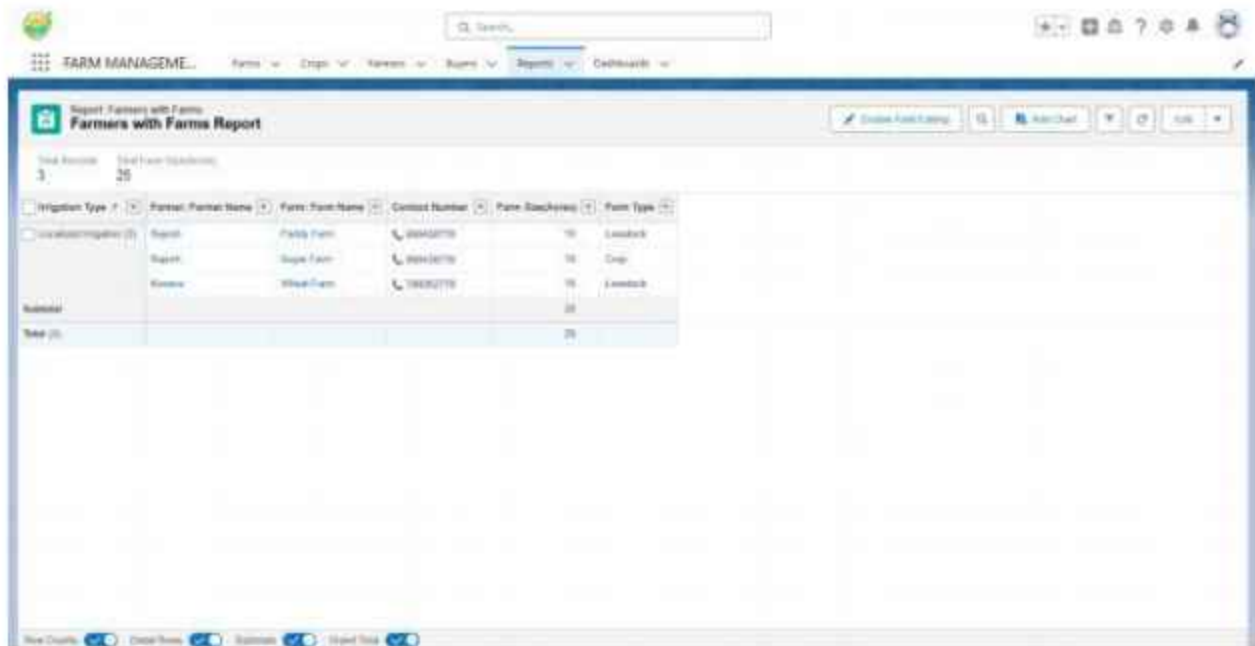
1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click +Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done.



Milestone-13: View Report and Dashboard:

Activity 1: Report

- Click on App Launcher on left side of screen.
- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records

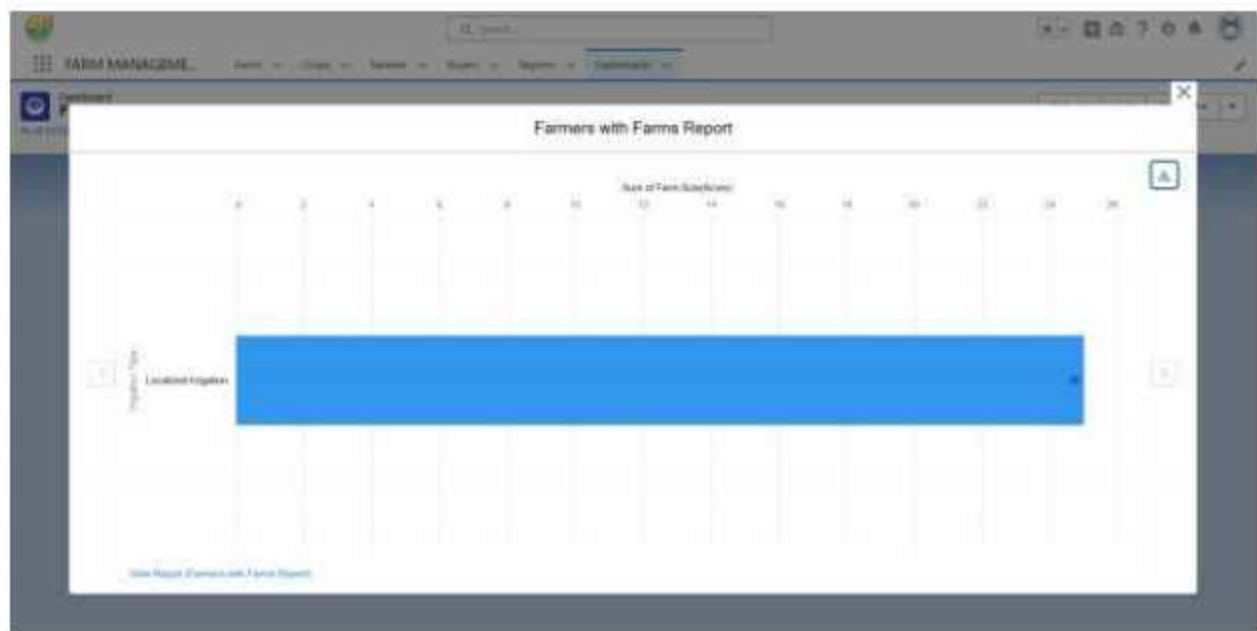


The screenshot shows the 'Farmers with Farms Report' page in the Farm Management System. The page includes a search bar, navigation tabs (Farm, Crop, Farmer, Report, Dashboard), and a table of reports. The table has columns for Report, Farm Name, Contact Number, Farm Location, and Farm Type. The data shows three reports for 'Farm Name' with contact numbers 099450770, 099450770, and 099450770, all located in 'Farm Location' and of type 'Farm'. A 'Total' row shows 3 reports and 25 farms.

Report	Farm Name	Contact Number	Farm Location	Farm Type
Report	Farm Name	099450770	Farm Location	Farm
Report	Farm Name	099450770	Farm Location	Farm
Report	Farm Name	099450770	Farm Location	Farm
Total				25

Activity 2: Dashboard

- Click on App Launcher on left side of screen.
- Search Farm Management system & click on it.
- Click on Dashboard Tab.
- Click Farmers With farms & see records.



7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce interface to verify system functionality. The following were tested:

- Creation of Farm, Crop, Farmer, and Buyer records
- Validation of input fields
- Relationship and dependency checks
- Profile and role access testing
- Sharing rule verification
- Report and dashboard generation

Performance testing ensured smooth system response and stability during multiple user operations.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data system
- Reduced paperwork
- Improved accuracy
- Better monitoring
- Real-time analytics

Disadvantages

- Requires internet connection
- Salesforce license dependency
- Limited offline access

9. Conclusion:

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

10. FUTURE SCOPE

- Mobile application integration
- AI-based crop prediction
- Weather data integration
- IoT sensor connectivity
- Government scheme integration
- Market price forecasting

11. APPENDIX

- Source Code: Salesforce Configuration and Metadata
- Dataset: Farm, Crop, Farmer, and Buyer Records
- Tools Used: Salesforce Developer Org

Thank You

